Instrumentation for structural load measurements - with reference to the V52 turbine

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Learning objects

After this lection the student will be able to:

- Explan what type instrumentation are used for structural measurements on the V52 turbine;

- Explain where the instrumentation are located on the V52 turbine;

- Identify the typical operational modes for the V52 turbine.

Literature

Please note that this course does not present the strain gauge theory and the Wheatstone bridge circuit. Further information about this topic can be found in the literature listed below:

1) Stefan Kiel: *Technology and Practical Use of Strain Gages.* Wilhelm Ernst & Sohn, 2017, Wiley.

2) Karl Hertz: *An Introduction to Measurements using Strain Gauges*. Hottinger Balwind Messtechnik GmbH, Darmstadt, book

3) Karl Hertz: *An Introduction to Stress Analysis using Strain Gauges*. HBM Test and Measurement, accessible from: www.hbm.com

4) Kart Hertz: *Practical hints for installing strain gauges.* Document s1421.pdf, HBM Test and Measurement, accessible from: www.hbm.com

Introduction

- Where and how to measure structural loads on a wind turbines are described in the IEC standard 61400-13.

- Lectures and exercises on this topic will be supplemented with real load measurements from a V52 wind turbine.

- The instrumentation for these load measurements on the V52 wind turbine are identified in this presentation:

    1) Structural loads measurements
    2) Operational parameters
    3) Meteorological measurements

- Identification of the operational sequences for the V52 turbine.

Basic instrumentation for structural measurements

- Strain gauges on the rotor
  - Mounted in the blades
  - Mounted on the main shaft

1) Flapwise blade bending
2) Edgewise blade bending

1) 2 x shaft bending
2) Shaft torque

- Strain gauges in the tower
  - Mounted in tower top
  - Mounted in tower bottom

Tower top torsion

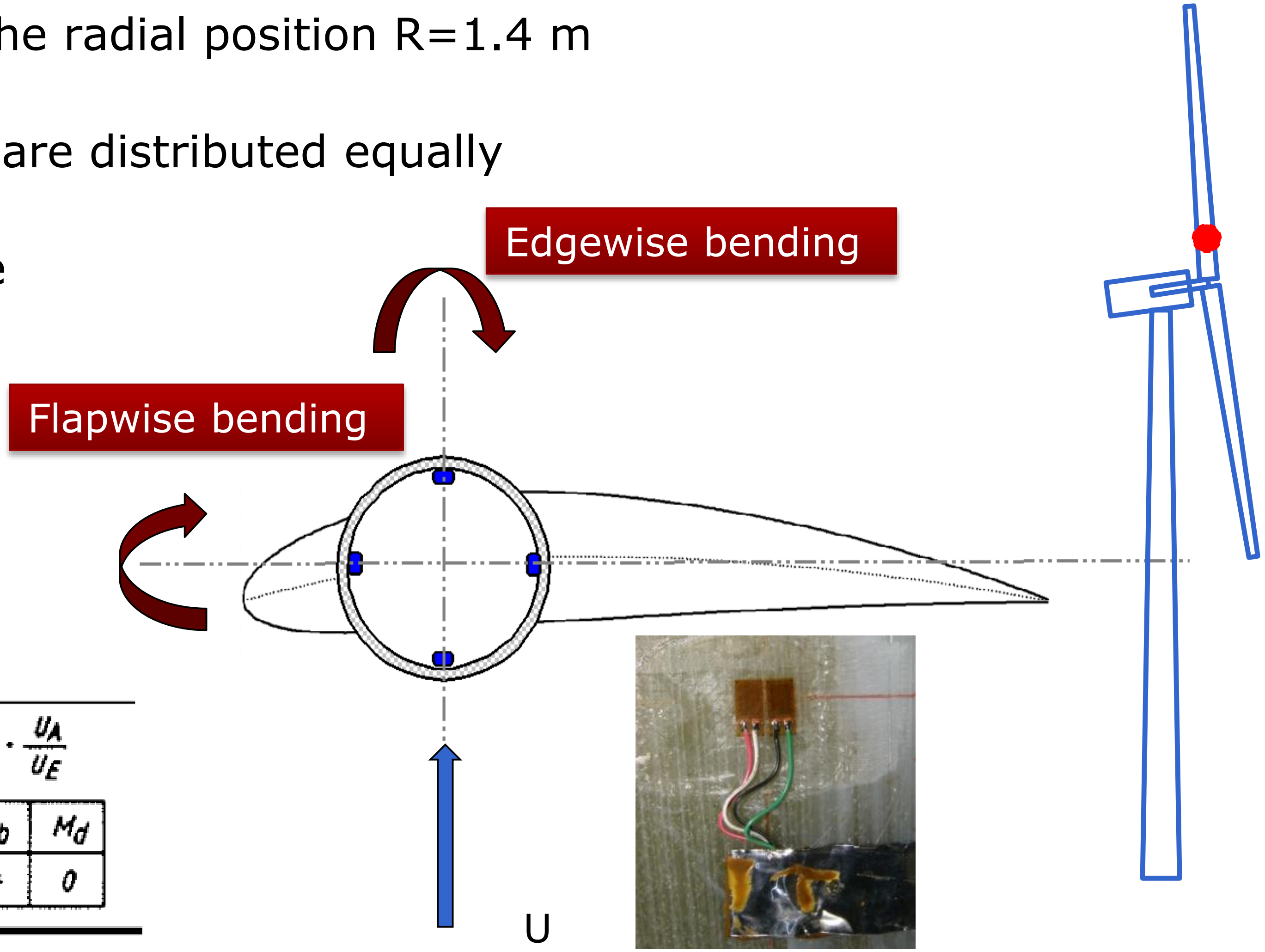
1) 2 x tower bottom bending

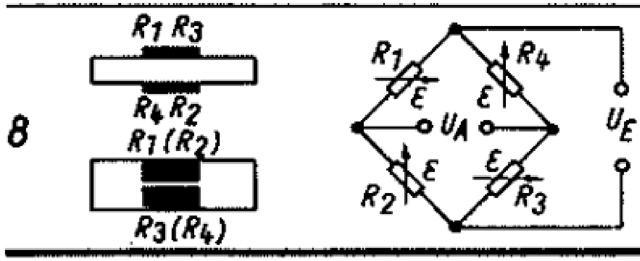
- Operational parameters
  - Pitch angle
  - Rotor speed – low speed & high speed shaft
  - Azimuth position
  - Active power
  - Yaw position

- Meteorological parameters (wind speed, direction,..)

Blade root bending moments – V52

- Strain gauge location is located at the radial position R=1.4 m in the circular section of the blade.
- 4 groups, each consisting of 2 SG's are distributed equally with an angular distance of 90 deg.
- Each of the two opposite groups are connected to a Wheatstone Bridge as shown on the previous slide.
- Output signals are µstrain in flapwise and edgewise direction respectively.
- Type: SG Bridge Hottinger No 8



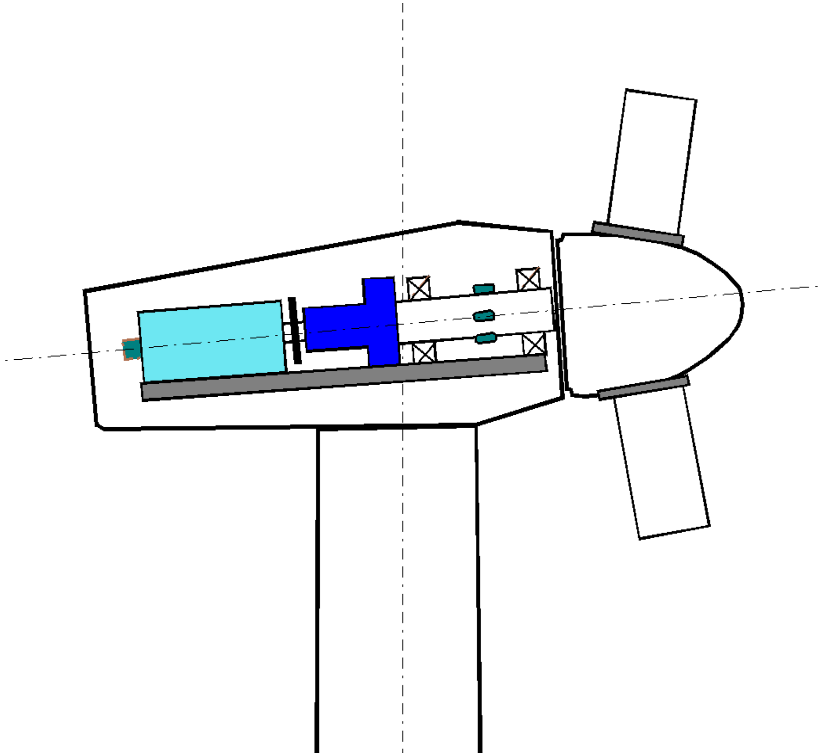


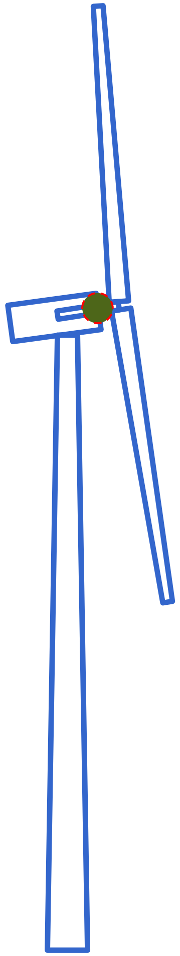
$$\varepsilon = \varepsilon_b = \frac{1}{4} \cdot \frac{4}{k} \cdot \frac{U_A}{U_E}$$

t	P	Mb	Md
0	0	4	0

Main shaft bending & torque – V52

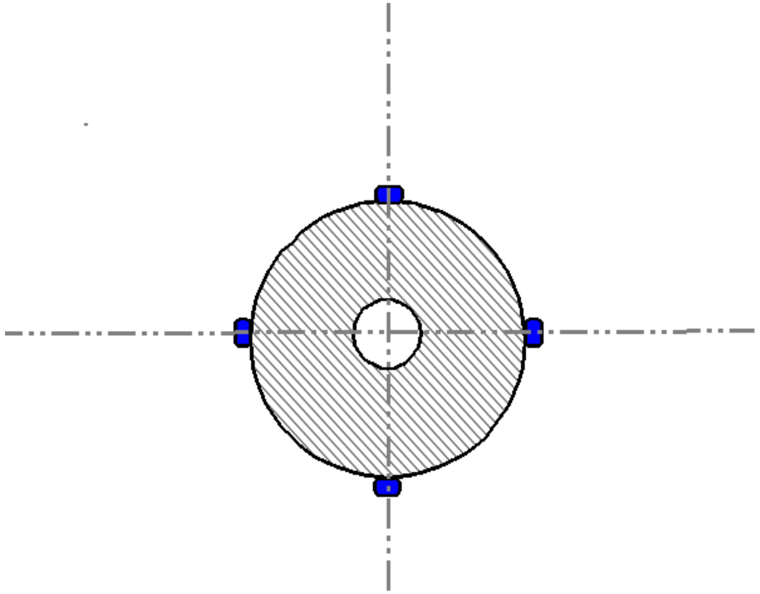
- Strain gauge location: main shaft, between main bearings.

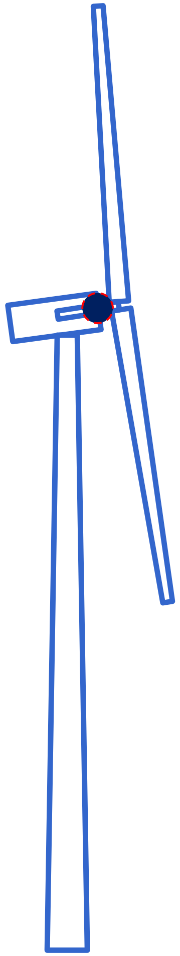


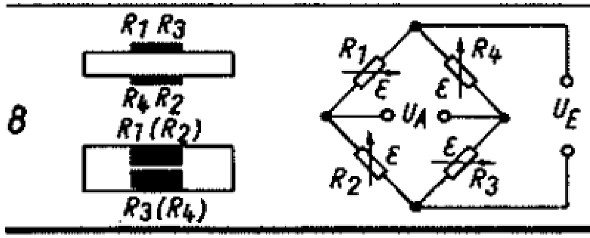


Main shaft bending– V52

- Strain gauge location is located on the main shaft surface.
- 4 groups, each consisting of 2 SG’s are distributed equally with an angular distance of 90 deg.
- Each pair consisting of two opposite groups are connected to a Wheatstone Bridge as shown on the previous slide.
- Output signals are μstrain in 0-180º and 90-270º directions respectively.
- Type: SG Bridge Hottinger No 8





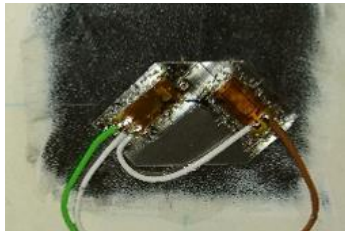


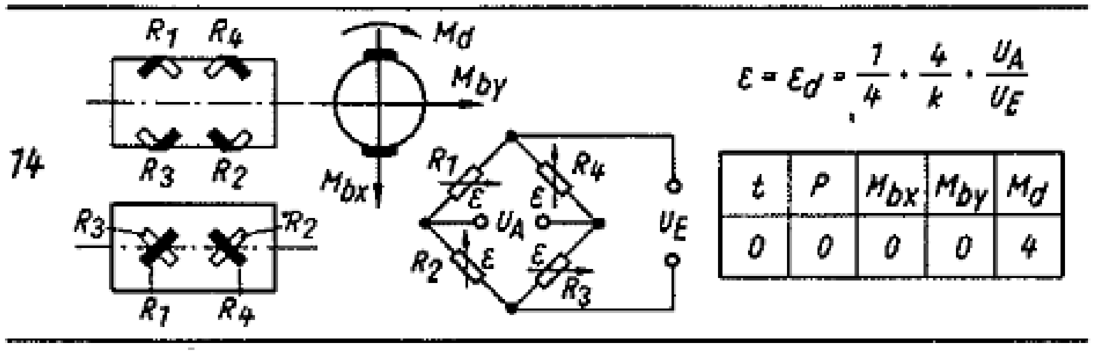
$$\varepsilon = \varepsilon_b = \frac{1}{4} \cdot \frac{4}{k} \cdot \frac{U_A}{U_E}$$

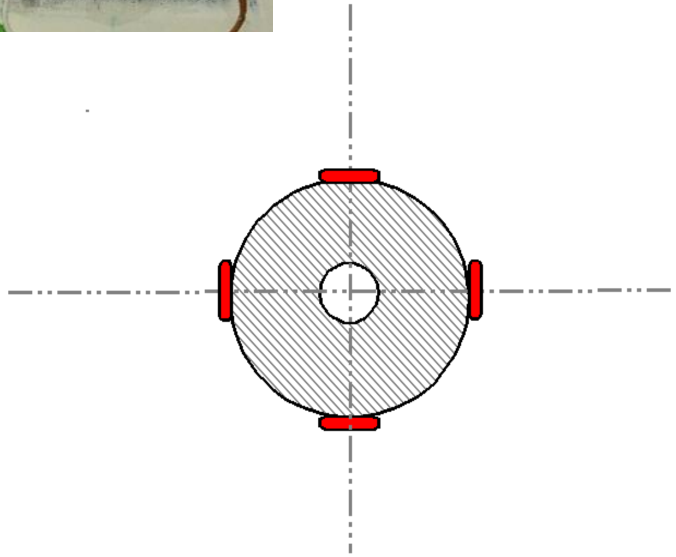
t	P	M_b	M_d
0	0	4	0

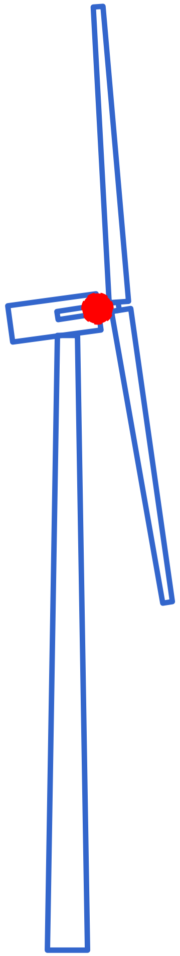
Main shaft torque – V52

- Torque is measured with 2 pairs of rosette SGs mounted 45º compared to shaft direction.
- Strain gauge location is located on the main shaft surface between the main bearings.
- 2 pairs of SG's are mounted with an angular distance of 180 deg, connected to one Wheatstone Bridge.
- Output signals is µstrain in in the shear direction.
- Type: SG Bridge Hottinger No 14



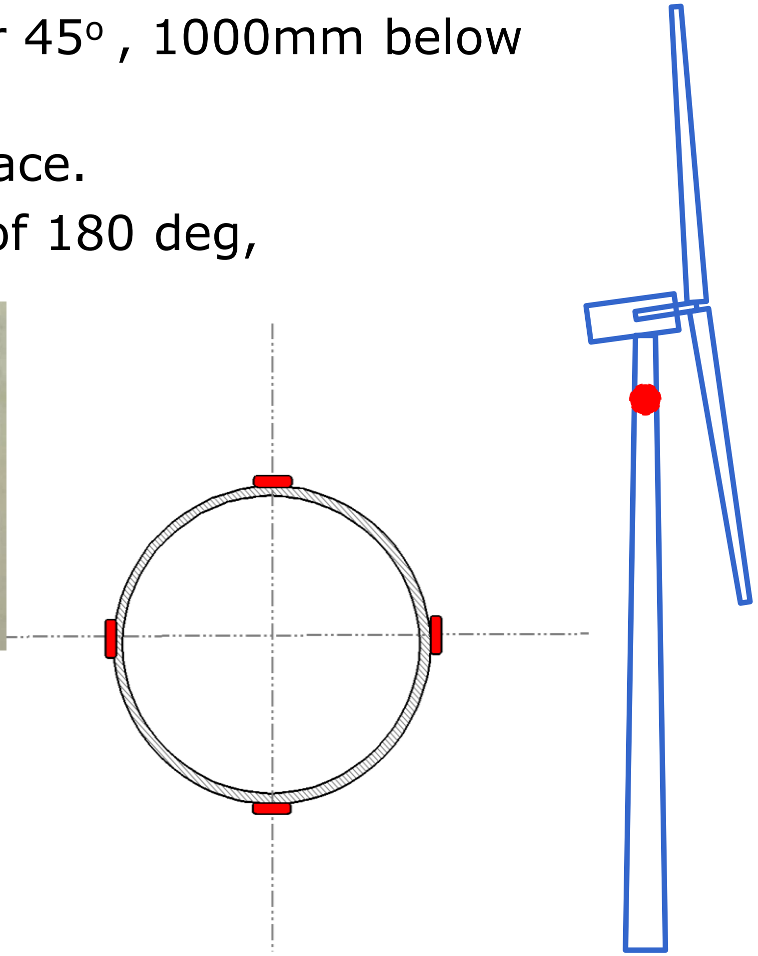


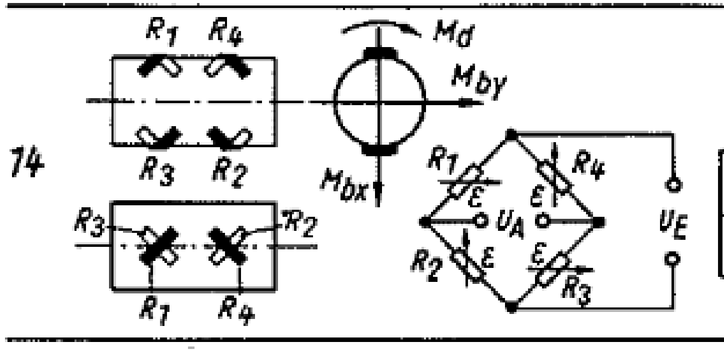




Tower top torsion – V52

- Torque is measured with 4 rosette SGs mounted under 45º , 1000mm below the tower top flange.
- Strain gauge location is located on the main shaft surface.
- 2 pairs of SG's are mounted with an angular distance of 180 deg, connected to one Wheatstone Bridge.
- Output signals is μstrain in in the shear direction.
- Type: SG Bridge Hottinger No 14
- Problem: difficult to calibrate.



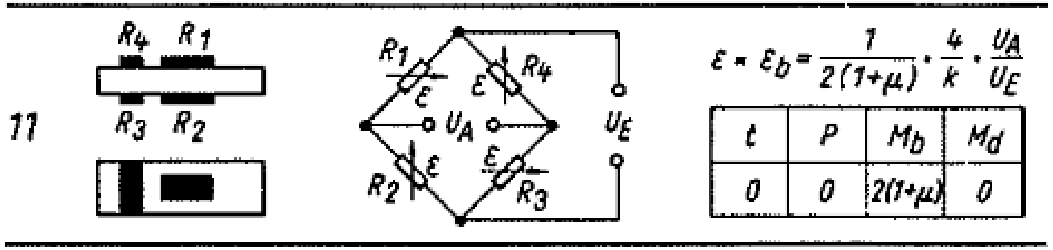


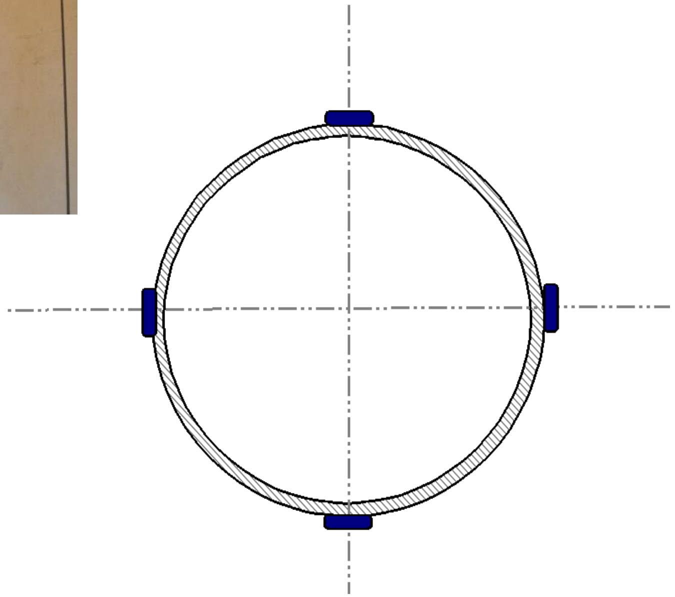
$$\varepsilon = \varepsilon_d \cdot \frac{1}{4} \cdot \frac{4}{k} \cdot \frac{U_A}{U_E}$$

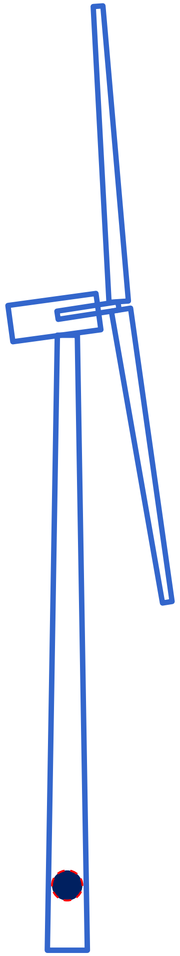
t	P	Mbx	Mby	Md
0	0	0	0	4

Tower bottom bending – V52

- Strain gauge location is located at height=2.5m above ground level in the circular section of the blade.
- 4 groups, each consisting of 2 SG's are distributed equally with an angular distance of 90 deg.
- Each of the two opposite groups are connected to a Wheatstone Bridge as shown on a previous slide.
- Output signals are µstrain in flapwise and edgewise direction respectively.
- Type: SG Bridge Hottinger No 11: bending with temperature compensation







Operational data

The operational data is either measured directly or provided by the wind turbine controller.

1) Rotor speed – low speed shaft (controller)

2) Rotor speed – high speed shaft (controller)

3) Pitch angle - (controller)

4) Rotor azimuth angle (measured)

5) Yaw position (controller)

6) Nacelle wind speed (controller)

2-D sonic nacelle wind speed – V52

Meteorological measurements

1) Wind speed at hub height, 44m

2) Wind direction below hub height, 41m

3) (Wind shear, range 18 – 70 m)

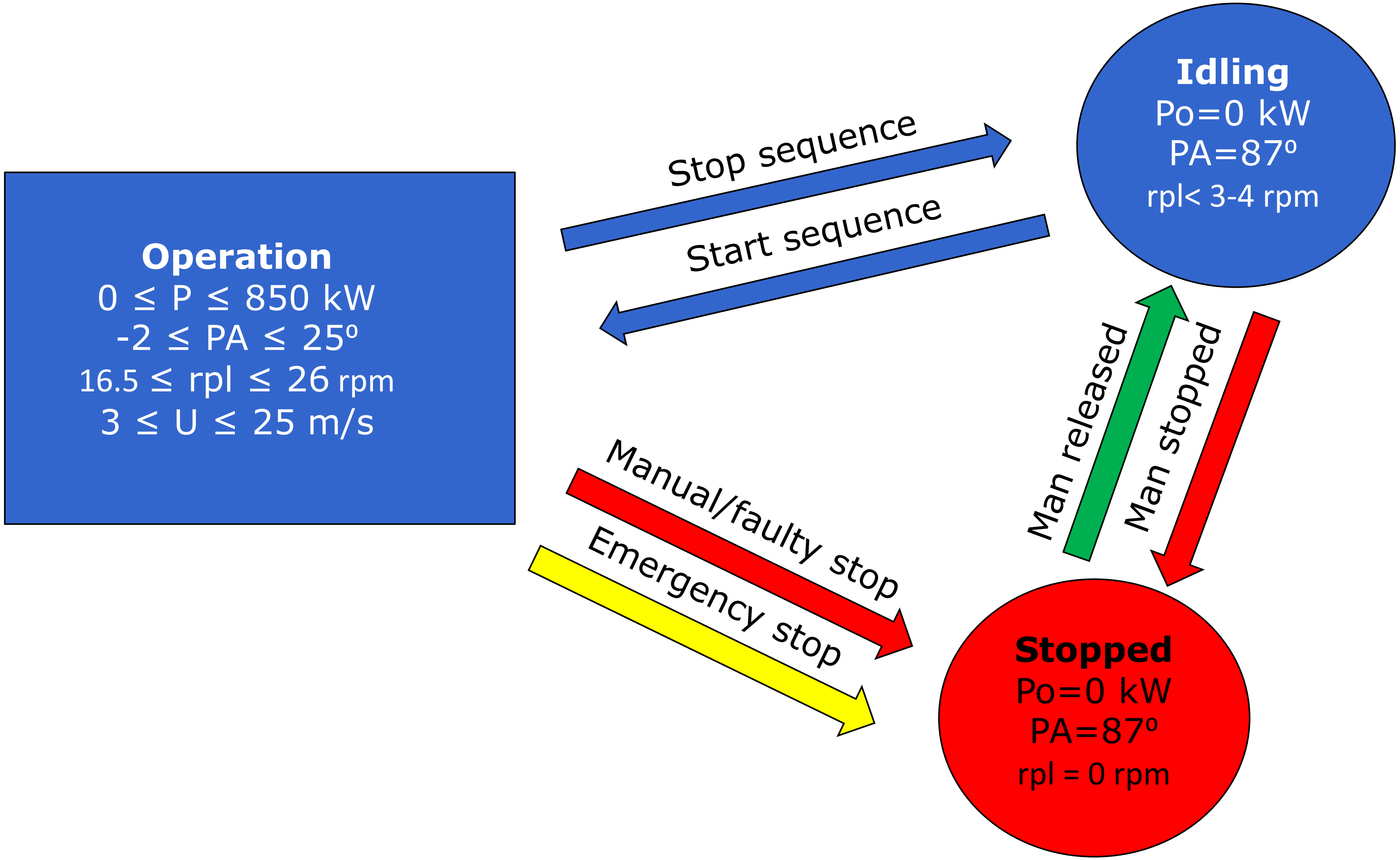
4) (Wind veer, range 18 – 70m)

5) Temperatures 18 & 70 m

6) Atmospheric Pressure – 2 m

7) Atmospheric stability, MO length, stability class

Operational wind turbine modes



Stop and start sequence – V52